

TERESA: Whole-Body Active Perception for Legged Robot-Assisted Emergency Assessment

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Abstract—Emergency assessment protocols such as Focused Assessment with Sonography in Trauma (FAST) ultrasound and the Airway, Breathing, Circulation, Disability, Exposure (ABCDE) survey are critical for rapid triage in mass-casualty incidents, disaster sites, and scenarios where trained clinicians are unavailable or overwhelmed. Deploying a legged mobile manipulator to autonomously perform these assessments could extend emergency care to inaccessible environments, but poses a fundamental challenge: the robot must actively reconfigure its body to compensate for perceptual uncertainty while navigating toward and scanning an unknown patient.

This paper presents Trustworthy Emergency Robot for Efficient Support and Assistance (TERESA) (Trustworthy Emergency Robot for Efficient Support and Assistance), an integrated system for autonomous emergency assessment mounted on a Boston Dynamics Spot legged robot equipped with a Unitree Z1 manipulator arm. The core of TERESA is a whole-body active perception framework in which the robot continuously adapts its posture—body height, pitch, and yaw—to maximise perceptual quality and bring anatomical targets into the arm's optimal workspace. A coarse-to-fine active search phase rotates the robot's body through a sequence of orientations, triggering a targeted refinement sweep whenever a camera detects a potential human form. Patient localisation is confirmed through a hybrid dual-sensor lock that combines a wide-field body camera with a close-range wrist camera before committing to navigation. An adaptive anticipatory scanner acquires multi-view anatomical data concurrently with approach motion, intelligently reducing the number of scanning poses when torso keypoints are already well resolved, thereby eliminating the idle time of traditional stop-then-scan approaches. During scanning, a pre-planned body reconfiguration optimises the robot's posture for each anatomical target through an offline search, maximising arm manipulability without requiring real-time whole-body optimisation.

Experiments on the physical Spot+Z1 platform demonstrate that TERESA achieves fully autonomous emergency assessment, with the active perception pipeline reducing mission idle time by **XX** % compared to a sequential stop-and-manipulate baseline while maintaining a probe positioning error below **XX** mm. To the best of our knowledge, this is the first legged robotic system to demonstrate perception-driven whole-body reconfiguration for autonomous emergency assessment.

Index Terms—Whole-body active perception, legged manipulation, medical robotics, emergency assessment, FAST ultrasound, mobile manipulation, body reconfiguration.

I. INTRODUCTION

EMERGENCY assessment protocols—including the Airway, Breathing, Circulation, Disability, Exposure (ABCDE) survey and Focused Assessment with Sonography in Trauma (FAST) ultrasound—enable clinicians to rapidly identify life-threatening conditions at the point of care. In mass-casualty incidents, disaster sites, and hazardous

environments, however, trained personnel are often unavailable or overwhelmed, creating a critical bottleneck in triage. A legged mobile manipulator that can autonomously navigate unstructured terrain, locate a patient, and perform these assessments could extend emergency care to scenarios where human responders cannot safely operate.

Autonomous robotic ultrasound has advanced significantly on fixed-base platforms. Systems employing impedance control, deep-learning-based segmentation, and pre-planned scan protocols have demonstrated clinically relevant probe positioning without direct human guidance [1]–[4]. These results, however, share a fundamental limitation: the patient must be pre-positioned within the arm's reachable workspace, confining autonomous ultrasound to controlled clinical settings.

Legged platforms such as the Boston Dynamics Spot remove this constraint. They can traverse stairs, rubble, and confined spaces inaccessible to wheeled robots, and can therefore reach patients wherever they lie. Mounting a manipulator arm on a legged base introduces, however, a deeper challenge that goes beyond motion coordination. The robot does not know the patient's location a priori, and the quality of its perception depends critically on how it positions its own body: camera viewpoint, height, and orientation all affect detection accuracy. Existing Whole-Body Control (WBC) formulations for legged manipulators [5]–[7] simultaneously optimise base and arm motion, but assume known or static targets and do not account for the coupling between body posture and perceptual quality. In a realistic emergency scenario, the robot must actively shape its own perception before it can act.

This paper presents Trustworthy Emergency Robot for Efficient Support and Assistance (TERESA) (Trustworthy Emergency Robot for Efficient Support and Assistance), an integrated system for autonomous emergency assessment on a Boston Dynamics Spot legged robot equipped with a Unitree Z1 manipulator arm. Building on the fixed-base scanning competence established in [4], we extend autonomous assessment to a fully mobile platform capable of operating in unstructured environments. The core contribution of TERESA is a whole-body active perception framework: the robot continuously adapts its posture—body height, pitch, and yaw—not merely to minimise kinematic error, but to maximise the quality of its perceptual input.

The system is built around four principles. First, a **hybrid confidence-gated active search** phase combines coarse body rotation with targeted refinement: the robot sweeps through orientations and, upon detecting a potential body with either of its two cameras, performs a focused sweep to maximise perceptual confidence before committing to a dual-sensor lock that confirms patient localisation. Second, an **adaptive**

anticipatory body scanner acquires multi-view anatomical data during base navigation, dynamically selecting the number of scanning poses based on how well the torso keypoints have already been resolved—sparse confidence triggers a more comprehensive sweep, while already-certain detections proceed with minimal scanning overhead. Third, a **perception-quality-aware velocity controller** scales the approach speed in proportion to detection confidence, avoiding both premature commitment to uncertain targets and unnecessary stops. Fourth, a **pre-planned body reconfiguration** step optimises the robot’s posture for each anatomical scanning target through an offline search, maximising arm manipulability without requiring real-time whole-body optimisation.

The main contributions of this paper are:

- 1) A **whole-body active perception framework** for legged mobile manipulation, in which body posture is optimised to improve perceptual quality rather than solely to minimise kinematic error (Section IV).
- 2) A **hybrid confidence-gated search strategy** that combines coarse rotation with targeted refinement and dual-sensor locking, ensuring robust patient localisation before the robot commits to navigation (Section IV).
- 3) An **adaptive multi-view scanning pipeline** that acquires patient anatomy concurrently with base motion and dynamically reduces scanning poses when keypoints are already well resolved, removing the idle periods inherent in sequential stop-and-scan architectures (Section VI).
- 4) **Experimental validation** on a physical Spot+Z1 platform, demonstrating fully autonomous emergency assessment with a **XX** % reduction in idle time compared to a stop-and-manipulate baseline (Section VIII).

To the best of our knowledge, TERESA is the first legged robotic system to demonstrate perception-driven whole-body reconfiguration for autonomous emergency assessment.

The remainder of this paper is organised as follows. Section II reviews related work in robotic ultrasound, legged whole-body control, and active perception for mobile manipulation. Section III formally defines the system and the autonomous emergency assessment problem. Section IV presents the active perception framework, including hybrid confidence-gated search and adaptive anticipatory scanning. Section V describes the decoupled whole-body reconfiguration approach. Section VI details the complete system architecture and mission Finite State Machine (FSM). Section VII describes the experimental setup and Section VIII presents the results and discussion. Section IX concludes the paper.

II. RELATED WORK

This section reviews the three main research threads that TERESA builds upon: autonomous robotic ultrasound, whole-body control for legged manipulators, and active perception for mobile manipulation. We conclude with a review of posture optimisation and body reconfiguration techniques for mobile manipulation.

A. Robotic Ultrasound Systems

Robot-assisted ultrasound has been an active research area for over two decades, driven by the need to reduce operator variability, enable remote diagnostics, and automate repetitive scanning procedures. Comprehensive surveys by von Haxthausen [8], Li et al. [2], and Munir et al. [9] document the evolution from early teleoperated systems to contemporary autonomous scanning platforms. Hidalgo et al. [10] review current clinical applications of robot-assisted ultrasound, highlighting both the progress and the persistent limitation that all deployed systems operate on fixed bases.

Two technical challenges have dominated the literature: probe positioning accuracy and contact force control. Gilbertson and Anthony [11] established a force/position decoupling framework for freehand ultrasound. Dyck et al. [12] extended impedance control to arbitrary curved surfaces using discrete differential geometry. Ning et al. [13] proposed force-guided scanning for soft tissue environments. Ma et al. [14] introduced an active-sensing end-effector that enables the probe to self-normalise to the skin surface—an early form of perception-driven adaptation.

On the planning side, Suligoj et al. [15] presented RobUSt, a system that autonomously acquires medical images following a pre-defined anatomical protocol. Duan et al. [3] demonstrated optimisation-based control for scoliosis assessment with variable impedance. In our own prior work [4], we showed that a collaborative robot arm on a fixed base can autonomously screen the abdominal aorta using impedance control and deep-learning-based segmentation.

A common thread across all these systems is the assumption that the patient is pre-positioned within the arm’s reachable workspace. No existing autonomous robotic ultrasound system has been deployed on a legged mobile platform capable of navigating to an unknown patient location in an unstructured environment. TERESA addresses this gap by mounting the ultrasound payload on a Boston Dynamics Spot quadruped and providing the full perception-to-scanning pipeline needed for autonomous operation.

B. Whole-Body Control for Legged Manipulators

WBC has emerged as the dominant paradigm for coordinating locomotion and manipulation on legged robots. Khatib [16] introduced the operational space formulation, and Yoshikawa [17] defined the manipulability measure that underpins singularity avoidance in most modern WBC approaches. Bayle et al. [18] extended manipulability analysis to wheeled mobile manipulators, showing that base motion can preserve arm dexterity.

A comprehensive survey by Wensing et al. [19] covers optimisation-based control for dynamic legged robots, including WBC and Model Predictive Control (MPC) formulations. For legged platforms with mounted arms, several hierarchical and holistic approaches have been proposed. Bellicoso et al. [5] demonstrated articulated locomotion and manipulation (ALMA) on a torque-controlled quadruped. Minniti et al. [20] formulated whole-body MPC for dynamically stable mobile manipulation, and Sleiman et al. [6] unified locomotion and

manipulation into a single MPC. Haviland et al. [7] proposed a holistic reactive WBC that simultaneously optimises arm and base velocities through a single Quadratic Program (QP). Li et al. [21] presented a hierarchical QP-based WBC for a torque-controlled quadrupedal manipulator, and Liu et al. [22] incorporated visual feedback into the WBC loop for loco-manipulation tasks. On the Boston Dynamics Spot platform, Zimmermann et al. [23] demonstrated dynamic grasping with an external arm, and Xie et al. [24] proposed adaptive motion distribution to enhance manipulability on legged manipulators.

Wolfslag et al. [25] studied optimisation of body-ground contact to augment whole-body loco-manipulation on quadruped robots, introducing the concept that body posture directly affects manipulation capability. This insight is central to TERESA's body reconfiguration strategy.

Despite these advances, all existing legged WBC formulations share a critical assumption: the manipulation target is known in the world frame a priori. They do not address the perceptual uncertainty inherent to detecting and approaching an unknown patient in an unstructured emergency scenario. TERESA extends WBC by closing the loop between perception quality and body posture: the robot actively reconfigures itself to improve detection, using WBC for reactive arm look-at during approach and pre-planned body reconfiguration for scanning.

C. Active Perception for Mobile Manipulation

Active perception—the principle that a robot should move to improve its sensory input—has been studied extensively in computer vision and robotics. Zeng et al. [26] provide a comprehensive survey of view planning in robot active vision, covering systems, algorithms, and applications. Bai et al. [27] survey information-driven path planning, where motion is optimised to maximise information gain from onboard sensors.

In the context of mobile manipulation, Zaky et al. [28] demonstrated that active perception strategies consistently outperform passive observation for goal-oriented manipulation tasks. Zhang and Scaramuzza [29] introduced the Fisher Information Field, a differentiable map that quantifies perceptual quality and can be integrated into motion planning. Tordesillas and How [30] proposed PANTHER, a perception-aware trajectory planner that optimises the vehicle's yaw to maintain visibility of dynamic obstacles, demonstrating the value of coupling perception constraints with motion optimisation.

Next-best-view planning has been applied to manipulation tasks. Breyer et al. [31] presented a closed-loop next-best-view planner for target-driven grasping, in which the robot continuously selects viewpoints that maximise information gain about the target object. Hu and Pagilla [32] proposed view planning for active robot perception using point clouds, optimising sensor poses for object pose estimation. Ren and Qureshi [33] developed active neural sensing and planning for robot manipulators operating in unknown cluttered environments, combining deep-learning-based scene understanding with viewpoint optimisation.

The closest works to our approach combine active perception with whole-body motion. Mittal et al. [34] demonstrated

whole-body mobile manipulation for articulated object interaction in unknown scenes, including experiments on legged platforms. Jauhri et al. [35] proposed active-perceptive motion generation for mobile manipulation, where a receding-horizon planner generates informative trajectories that balance task progress with perceptual quality. Xie et al. [36] introduced coupled active perception and manipulation planning for a mobile manipulator in agricultural settings, explicitly constraining the robot's state by perceptual requirements. Qin et al. [37] recently proposed a one-shot multimodal active perception framework that learns to predict optimal viewpoints for robotic manipulation.

However, none of these systems address the specific challenge of whole-body reconfiguration on a legged platform for medical tasks. Existing active perception frameworks operate on wheeled bases (Mittal et al., Jauhri et al.) or fixed manipulators (Ren and Qureshi), where the perception-action loop is limited to camera viewpoint or arm pose. On a legged platform, body height, pitch, and yaw can all be controlled to reshape the perceptual field—a richer action space that TERESA exploits through confidence-gated search and pre-planned body reconfiguration.

D. Posture Optimisation and Body Reconfiguration

Optimising the base pose of a mobile manipulator to maximise task performance has been studied in the context of reachability and manipulability. Makhal and Goins [38] introduced Reuleaux, a method for robot base placement via reachability analysis, generating maps of feasible end-effector positions as a function of base location. Fan et al. [39] proposed base position optimisation for machining large components, demonstrating that pre-planned base placement significantly improves manipulation quality. Zhao et al. [40] presented B*, an efficient optimal base placement algorithm for fixed-base manipulators.

Gong et al. [41] reviewed legged robots for object manipulation, identifying body posture optimisation as a key open challenge. Wolfslag et al. [25] explicitly optimised body-ground contact configurations on a quadruped to enhance whole-body loco-manipulation, showing that even small adjustments in body height and orientation can substantially affect arm reachability.

TERESA extends these ideas by embedding posture optimisation within an active perception loop. Rather than optimising base placement for a known task, the system first uses body posture to improve perceptual quality (Section IV), and then pre-plans body reconfiguration to bring each anatomical scanning target into the arm's optimal workspace (Section V). This two-stage approach allows the legged platform to serve both as a perception aid and as a manipulation enabler, closing the gap between active perception and whole-body reconfiguration for autonomous medical assessment.

III. PROBLEM FORMULATION

A. System Model

The TERESA platform consists of a Boston Dynamics Spot quadruped and a Unitree Z1 six-Degree of Freedom

(DOF) serial manipulator arm rigidly mounted on the robot's back. Four reference frames are defined: the world frame $\mathcal{F}_W$ (coincident with Spot odometry), the Spot body frame $\mathcal{F}_B$ (origin at the geometric centre of the base), the arm base frame $\mathcal{F}_0$ (coincident with Z1 link 0, fixed relative to $\mathcal{F}_B$), and the end-effector frame $\mathcal{F}_{EE}$ (at the Z1 flange, where the ultrasound probe is attached).

1) *Spot base kinematics*: The Spot base is modelled as a non-holonomic vehicle constrained to move on a planar surface. Its configuration is described by $\xi = [x, y, \theta]^\top \in \mathbb{R}^2 \times \mathbb{S}^1$, where (x, y) is the position of $\mathcal{F}_B$ in $\mathcal{F}_W$ and θ is the yaw angle. The non-holonomic constraint restricts the admissible base velocities to

$$\dot{\xi} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos \theta \\ \sin \theta \\ 0 \end{bmatrix} v_x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \omega_z, \quad (1)$$

where $v_x \in \mathbb{R}$ is the forward linear velocity and $\omega_z \in \mathbb{R}$ is the yaw angular velocity, both expressed in $\mathcal{F}_B$.

2) *Body posture*: In addition to planar navigation, Spot can reconfigure its body posture by changing its height $h \in \mathbb{R}$ and pitch angle $\varphi \in \mathbb{R}$ relative to the nominal standing configuration. The height h is a vertical offset of $\mathcal{F}_B$ (negative values lower the body, bringing the onboard camera closer to the ground); the pitch φ is a rotation about the body y -axis (positive values tilt the body forward, orienting the camera toward the floor). These parameters are commanded through the native `body_pose` interface of the Spot driver. The combined base configuration is therefore $[x, y, \theta, h, \varphi]^\top$.

3) *Z1 arm kinematics*: The Z1 arm has $n = 6$ revolute joints with configuration vector $\mathbf{q} = [q_1, \dots, q_6]^\top \in \mathbb{R}^6$ and joint velocity vector $\dot{\mathbf{q}} \in \mathbb{R}^6$. The forward kinematics map $\mathbf{T}_{EE}(\mathbf{q}) \in SE(3)$ gives the homogeneous transformation from $\mathcal{F}_0$ to $\mathcal{F}_{EE}$ via the standard product of exponentials. The arm geometric Jacobian $\mathbf{J}_{\text{arm}}(\mathbf{q}) \in \mathbb{R}^{6 \times 6}$ relates joint velocities to the spatial velocity of $\mathcal{F}_{EE}$ expressed in $\mathcal{F}_0$:

$$\mathbf{v}_{EE}^{\text{arm}} = \mathbf{J}_{\text{arm}}(\mathbf{q}) \dot{\mathbf{q}}. \quad (2)$$

4) *Unified control input*: During the approach phase, the system commands the arm joints and the base velocity simultaneously through the combined input

$$\mathbf{u} = \begin{bmatrix} \dot{\mathbf{q}} \\ v_x \\ \omega_z \end{bmatrix} \in \mathbb{R}^8, \quad (3)$$

subject to the velocity bounds

$$\mathbf{u}_{\min} \leq \mathbf{u} \leq \mathbf{u}_{\max}, \quad (4)$$

where $\mathbf{u}_{\min}, \mathbf{u}_{\max} \in \mathbb{R}^8$ encode joint velocity limits for the arm and maximum linear and angular speed for the base. Body posture commands (h, φ) are not part of the continuous control input; they are set at discrete instants between mission phases, with the robot given a settle time before the next phase begins.

B. Perception Model

The quality of patient detection depends on the robot's body posture, as the onboard Orbbec RGB-D (RGBD) camera is

rigidly attached to $\mathcal{F}_B$. Let $c(t) \in [0, 1]$ denote the detection confidence provided by the You Only Look Once (YOLO)-based perception pipeline at time t . We do not assume an analytical form for the mapping from posture to confidence, but we rely on the empirical observation that body height and pitch significantly affect the camera's field of view and, consequently, the reliability of patient localisation. The system operates under the premise that there exists a posture (h^*, φ^*) that maximises detection quality for a given scene geometry, and that this posture can be discovered through active search.

During scanning, the anatomical target positions are estimated in the odometry frame $\mathcal{F}_W$ by the anticipatory body scanner (Section IV). These estimates are transformed into the arm base frame $\mathcal{F}_0$ at execution time via the Transform (TF) tree, so that changes in body posture (h, φ) are automatically reflected in the Inverse Kinematics (IK) goal without requiring re-estimation of the target.

C. Task Definition

An autonomous emergency assessment mission requires the End-Effector (EE) to reach a sequence of N anatomical target poses. Let $\mathcal{T} = \{\mathbf{T}_i^* \in SE(3)\}_{i=1}^N$ denote the desired EE poses corresponding to the anatomical windows of interest (e.g., the four FAST quadrants). Each target pose $\mathbf{T}_i^* = (\mathbf{R}_i^*, \mathbf{p}_i^*)$ comprises a desired EE position $\mathbf{p}_i^* \in \mathbb{R}^3$ and orientation $\mathbf{R}_i^* \in SO(3)$, both expressed in $\mathcal{F}_W$.

The target poses are not known a priori. They are acquired in two stages. During the approach phase, a wide-field patient detection pipeline provides a time-varying estimate of the patient's torso position $\hat{\mathbf{p}}(t)$ in the camera frame, which the system transforms to $\mathcal{F}_W$ once the patient has been localised with sufficient confidence. During the approach and early scanning phases, an anticipatory multi-view body scanner maps the patient's anatomy in $\mathcal{F}_W$, from which the target set $\mathcal{T}$ is computed.

D. Problem Statement

Given the system model defined in (1)–(4) and the unknown patient position, the autonomous emergency assessment problem is to find a control sequence $\mathbf{u}(t)$ and a sequence of body posture commands $(h(t), \varphi(t))$ that enable the robot to:

- 1) **Active perception**: discover a body posture $(h^*, \varphi^*, \theta^*)$ from which detection confidence exceeds a threshold $c_{\min}$, ensuring robust patient localisation before committing to the approach;
- 2) **Navigation**: drive the Spot base from its initial pose to a configuration from which all targets in $\mathcal{T}$ lie within the reachable workspace of the Z1 arm, using the online patient estimate $\hat{\mathbf{p}}(t)$;
- 3) **Body reconfiguration**: for each scanning target $i = 1, \dots, N$, select a body posture (h_i, φ_i) that brings the target as close as possible to a preferred workspace centre $\mathbf{p}_{\text{sweet}} \in \mathbb{R}^3$ in $\mathcal{F}_0$, maximising arm manipulability at execution time;
- 4) **Probe placement**: bring the EE to each target pose $\mathbf{T}_i^*$ in sequence, satisfying

$$\|\mathbf{p}_{EE}(t_i) - \mathbf{p}_i^*\| \leq \varepsilon_p, \quad (5)$$

where $\mathbf{p}_{EE}$ is the current EE position and $\varepsilon_p > 0$ is the admissible position error;

- 5) **Input feasibility**: satisfy the velocity bounds in (4) and the body posture constraints

$$h_{\min} \leq h \leq h_{\max}, \quad 0 \leq \varphi \leq \varphi_{\max}, \quad (6)$$

at all times.

The following assumptions are adopted throughout this paper:

- A1 The patient remains stationary during the scanning phase.
- A2 Spot operates on flat terrain, so the non-holonomic model (1) is valid.
- A3 The rigid transform between $\mathcal{F}_B$ and $\mathcal{F}_0$ is known and constant (fixed mount).
- A4 A single patient is present in the scene.
- A5 The onboard cameras are the only source of perception; no external infrastructure (e.g., external tracking systems) is available.

IV. WHOLE-BODY ACTIVE PERCEPTION

The core contribution of TERESA is a whole-body active perception framework: the robot continuously adapts its posture—body height, pitch, and yaw—to maximise the quality of its perceptual input. This section describes the three mechanisms that realise this principle: hybrid confidence-gated active search (Section IV-A), adaptive anticipatory body scanning with WBC (Section IV-B), and perception-quality-aware velocity control (Section IV-C).

A. Hybrid Confidence-Gated Active Search

Before the robot can approach the patient, it must first localise them reliably. In an unstructured emergency scenario, the patient’s position is unknown a priori, and the detection quality of the onboard Orbbec RGBD camera depends strongly on the robot’s body posture: a higher camera provides a wider field of view but reduces resolution at the ground plane; a forward pitch directs the camera toward the expected patient location but narrows the horizon. TERESA treats this coupling as an opportunity rather than a limitation: the robot actively searches over body postures to find the configuration that maximises detection confidence.

1) *Coarse rotation with targeted refinement*: The robot enters a **SEARCHING** phase in which it sweeps through a sequence of body yaw orientations at a nominal pitch, pausing at each to allow both cameras to observe the environment. A second camera—an EE-mounted RealSense—runs a torso tracker in a lenient guidance mode, capable of detecting any human keypoint without requiring full torso visibility. At each orientation, two conditions are evaluated concurrently:

- **Orbbec direct detection**: if the wide-field camera detects a lying patient with high posture confidence and a valid approach point can be computed, the robot transitions immediately to target locking—this is the dominant path when visibility is favourable.
- **RealSense guidance**: if the RealSense tracker detects a potential body (even a partial set of keypoints) while the Orbbec has not yet confirmed, the robot enters a

semi-locking phase. It computes the optimal yaw and pitch to align the Orbbec with the detected body and rotates the base accordingly, while the arm maintains a continuous look-at orientation toward the target to keep the EE camera on the region of interest. The Orbbec receives a clean observation window of several seconds to confirm the detection.

If neither condition is met after the coarse sweep completes, the robot triggers a **refinement phase**: at the most promising orientation (where either camera reported above-threshold confidence), it performs a targeted sweep of body pitch angles, maximising the Orbbec confidence score. If the best pitch yields confirmation, the lock proceeds; otherwise the coarse search resumes from the next orientation.

This two-tier strategy combines the speed of a sparse coarse search with the precision of targeted refinement, avoiding the idle time of an exhaustive posture grid while ensuring that narrow detections are not missed.

2) *Dual-sensor confidence lock*: Patient localisation is confirmed through a hybrid lock that fuses information from both cameras. When the Orbbec detects a lying posture with high confidence and a valid approach point, the robot enters a **LOCKING** state. The arm returns to an elevated home pose, providing the EE-mounted RealSense with an optimal vantage point for fine-grained torso analysis. The coordinator simultaneously collects a short sequence of approach point estimates from the Orbbec, transforms them to the world-fixed odometry frame $\mathcal{F}_W$ via the TF tree, and computes their mean as the initial approach target. The transition to the approach phase is gated by both the completion of the arm’s home trajectory (confirmed by the IK solver) and the collection of the required number of samples—ensuring that the robot is mechanically prepared and the target is statistically stable before committing to navigation.

This dual-sensor architecture provides complementary coverage: the Orbbec offers wide-field posture classification and coarse localisation, while the RealSense guides fine base alignment when the body is partially visible and confirms the target at close range during the subsequent pre-approach phase.

3) *Robustness to confidence loss*: If the Orbbec detection is temporarily lost during sample collection—for instance, due to a transient occlusion or a change in lighting—the system tolerates a brief absence before abandoning the lock and resuming the coarse search from the current orientation. This closed-loop behaviour prevents commitment to a stale or degraded target without reacting to momentary flickering.

Once the target is locked and the arm is in its home pose, the robot straightens its body posture (returning to nominal height and zero pitch), and the mission transitions to the pre-approach phase. During pre-approach, the EE-mounted RealSense camera becomes the primary sensor: the arm maintains a WBC look-at orientation toward the patient’s torso centroid (computed from the Orbbec skeleton keypoints during locking), and the coordinator waits for consecutive tracker confirmations before authorising the final approach.

B. Adaptive Anticipatory Body Scanning with Whole-Body Control

In traditional mobile manipulation pipelines, the robot first navigates to a target position, stops, and only then begins visual scanning—a sequential approach that introduces significant idle time. TERESA eliminates this idle period by executing the body scan concurrently with the approach navigation. This anticipatory scanning is enabled by a decoupled architecture in which the Spot base navigates toward the patient while the Z1 arm simultaneously performs a multi-view scan of the patient’s torso.

1) *Decoupled whole-body coordination*: During the approach phase, two independent controllers run in parallel. The Spot base is driven by a P-controller that computes forward velocity $v_x = k_p^{\text{lin}} \Delta d$ and yaw rate $\omega_z = k_p^{\text{ang}} \Delta \theta$ from the distance and angle error between the current body pose and the target in $\mathcal{F}_W$. The arm is controlled by a WBC QP that computes joint velocities $\dot{q}$ via a damped pseudo-inverse of the arm Jacobian, generating an EE orientation that points the RealSense camera toward the patient’s torso centroid. The WBC is used here not merely to optimise kinematics, but to serve the active perception objective: it continuously reorients the sensor to maximise the quality of the anatomical data being acquired.

2) *Adaptive Cartesian scan grid*: The arm executes a perceptual body scan using a dynamically generated Cartesian grid of poses around the current EE position. Unlike a fixed scan pattern, the grid adapts to the pre-existing torso detection quality. During the pre-approach phase, the RealSense tracker publishes per-keypoint confidence scores for the four canonical torso landmarks (left and right shoulders, left and right hips). These scores drive the grid generation:

- If all four torso keypoints are detected with high confidence, the scan uses a minimal grid: the current EE pose (already oriented toward the torso via the WBC look-at controller), advanced forward toward the patient, and a second elevated viewpoint for depth cross-validation. This two-pose scan completes rapidly, as the torso is already well resolved.
- If any keypoint falls below the confidence threshold, the grid expands to a comprehensive pattern: the base pose plus symmetric lateral offsets and an elevated frontal viewpoint, with the home pose interleaved between each lateral transit to ensure smooth IK trajectories. This four-pose sweep guarantees multi-angle coverage of the torso region, recovering keypoints that were partially occluded or at the periphery of the camera frustum.

All scanning poses include a forward advance toward the patient, shifting the camera closer to the torso for improved depth resolution. Each pose maintains a minimum-rotation look-at orientation that points the camera optical axis toward the target while preserving the wrist configuration, and is clipped against the arm’s reachable workspace with a safety margin.

3) *Multi-view fusion and FAST target generation*: At each pose, the BodySearchScanner module fuses the RealSense depth and skeleton data into 3D keypoint estimates of the

patient’s torso landmarks (shoulders, hips, sternum). The per-pose data collection runs for a fixed interval, with early termination if a high-quality detection is achieved before the timeout. After all poses are visited, the multi-view estimates are fused through a weighted average that accounts for detection rate, average confidence, and 3D spatial stability. From the fused torso model, the five canonical FAST anatomical target poses (Hub, Subxiphoid, RUQ, LUQ, Suprapubic) are computed as fixed offsets from the estimated torso centroid, expressed in the arm base frame $\mathcal{F}_0$.

The adaptive nature of the grid yields two key advantages. First, it reduces the total scan duration when visibility is favourable—the majority of cases—without compromising anatomical localisation when keypoints are partially occluded. Second, it eliminates the need for a separate phase-three conditional refinement step: the grid itself encodes the logic of “comprehensive when uncertain, minimal when confident.”

4) *Z1 FSM integration*: Once the scan is complete and the FAST targets are published in $\mathcal{F}_0$, a ready signal is broadcast. The Z1 FSM, which normally executes its own body scanning routine, receives this signal and skips its internal scan phase, transitioning directly to the scanning sequence. This integration eliminates redundant scanning and ensures that the body scan performed during approach is directly leveraged for the subsequent ultrasound acquisition.

C. Perception-Quality-Aware Velocity Control

During the approach, the quality of the patient detection can vary due to partial occlusions, changes in body posture, or motion of the platform. A fixed approach velocity would be either overly conservative (taking unnecessary time when detection is certain) or unsafe (charging toward an uncertain target). TERESA addresses this trade-off through a QualityMonitor that continuously maps detection confidence to approach velocity.

1) *Quality monitor*: The QualityMonitor maintains a target position in $\mathcal{F}_W$ and an associated quality value $q \in [q_{\min}, q_{\max}]$ measured in metres. The target is initialised from the confidence-locked sample mean obtained during the active search phase. It is subsequently updated only if a new detection arrives with confidence that exceeds the current best confidence by a minimum margin, preventing the target from drifting toward lower-quality measurements. The quality value is computed from the posture confidence c as:

$$q = q_{\max} (1 - c), \quad (7)$$

where $q_{\max}$ is a ceiling parameter. When confidence is high ($c \rightarrow 1$), the quality is low (small uncertainty radius); when confidence drops, the quality grows, reflecting increasing positional uncertainty. If no new confidence data arrive—for instance, during a temporary occlusion—the quality grows linearly with time, modelling the accumulation of uncertainty in the absence of new observations:

$$q(t + \Delta t) = q(t) + \dot{q}_{\text{growth}} \Delta t. \quad (8)$$

2) *Velocity scaling*: The base approach velocity is scaled by a smooth function of the current quality:

$$v_{\text{scale}}(q) = v_{\min} + \frac{1 - v_{\min}}{1 + q/q_{\text{ref}}}, \quad (9)$$

where $v_{\min} > 0$ is a floor that ensures the robot never stops completely (avoiding deadlocks when quality is high), and q_{ref} is a reference quality at which the velocity is reduced to $(1 + v_{\min})/2$. The scaling is applied multiplicatively to both the forward velocity v_x and the yaw rate ω_z :

$$v_x^{\text{cmd}} = v_{\text{scale}}(q) v_x, \quad \omega_z^{\text{cmd}} = v_{\text{scale}}(q) \omega_z. \quad (10)$$

When detection is reliable ($q \approx q_{\min}$, i.e., high confidence), $v_{\text{scale}} \approx 1$ and the robot approaches at full speed. As confidence degrades (q grows), the velocity scales down smoothly, giving the perception pipeline more time to re-acquire the target. When no new detections arrive, the linear growth of q in (8) produces a gradual deceleration, ensuring that the robot does not charge blindly toward a lost target.

The QualityMonitor thus closes the perception-action loop: the robot's motion is continuously modulated by the quality of its own perceptual input, embodying the principle that action should be proportional to certainty.

V. WHOLE-BODY RECONFIGURATION AND CONTACT CONTROL

The TERESA system employs three complementary control strategies that are active in different phases of the mission. During the approach, a WBC formulation drives the arm to maintain an active look-at orientation toward the patient, feeding the anticipatory body scanner. Between scanning targets, a pre-planned body reconfiguration step optimises the legged base posture to bring each anatomical target into the arm's optimal workspace. During scanning, a Cartesian impedance controller—built specifically for the Z1 arm, which lacks native impedance hardware—maintains compliant probe contact on the patient's skin. The handoff between these controllers is managed by the mission FSM described in Section VI.

A. Whole-Body Control for Reactive Arm Look-At

During the approach phase, the Z1 arm must continuously orient its EE-mounted RealSense camera toward the patient's torso to feed the anticipatory body scanner (Section IV-B). The Spot base simultaneously navigates toward the target fixed in the odometry frame $\mathcal{F}_W$. TERESA uses a WBC formulation in which the arm joint velocities are computed to track a desired EE orientation, while the base velocity is scaled by the perception-quality monitor.

1) *Arm look-at via damped pseudo-inverse*: Let $\mathbf{p}_{EE} \in \mathbb{R}^3$ be the current EE position and $\mathbf{p}^* \in \mathbb{R}^3$ the target position, both expressed in $\mathcal{F}_W$. The desired EE orientation is computed so that the EE x -axis points toward the target. Using the current arm configuration $\mathbf{q}$ and the arm Jacobian $\mathbf{J}_{\text{arm}}(\mathbf{q}) \in \mathbb{R}^{6 \times 6}$ defined in (2), the joint velocity command $\dot{\mathbf{q}} \in \mathbb{R}^6$ is obtained via a damped pseudo-inverse:

$$\dot{\mathbf{q}} = \mathbf{J}_{\text{arm}}^\top (\mathbf{J}_{\text{arm}} \mathbf{J}_{\text{arm}}^\top + \mu \mathbf{I}_6)^{-1} \mathbf{v}_{\text{des}}, \quad (11)$$

where $\mu > 0$ is a damping factor that ensures numerical stability near kinematic singularities. The desired spatial velocity $\mathbf{v}_{\text{des}} \in \mathbb{R}^6$ is:

$$\mathbf{v}_{\text{des}} = \begin{bmatrix} k_r \mathbf{e}_{\text{rot}} \\ \mathbf{0}_3 \end{bmatrix}, \quad (12)$$

where $k_r > 0$ is a rotational gain and $\mathbf{e}_{\text{rot}} \in \mathbb{R}^3$ is the orientation error between the current EE orientation and the desired look-at orientation. The orientation error is computed using a minimum-rotation formulation: given the home EE orientation $\mathbf{R}_{\text{home}} \in SO(3)$ and the desired look-at direction $\mathbf{x}_{EE} = (\mathbf{p}^* - \mathbf{p}_{EE})/\|\mathbf{p}^* - \mathbf{p}_{EE}\|$, the desired rotation is the quaternion that maps $\mathbf{R}_{\text{home}} \mathbf{e}_x$ to $\mathbf{x}_{EE}$ with minimum angular displacement, relaxing the wrist configuration. The translational component of $\mathbf{v}_{\text{des}}$ is zero because the translational target during scanning is provided separately by the body scanner (Section IV-B); during approach, the arm maintains a nominal home position and contributes only orientation.

2) *Base velocity modulation*: The Spot base velocity, consisting of forward velocity v_x and yaw rate ω_z , is controlled using a proportional law on the distance and angle to the target in $\mathcal{F}_W$. These raw commands are then scaled by the quality-aware velocity factor $v_{\text{scale}}(q)$ derived from the QualityMonitor in (9), as per (10). The base velocity is thus continuously modulated by perceptual confidence, closing the whole-body perception-action loop.

3) *Robustness to TF degradation*: In a distributed system spanning the Spot onboard computer and an external PC, the TF tree can experience transient failures due to clock desynchronisation or Data Distribution Service (DDS) discovery delays. To maintain continuity of base motion, the controller caches the last valid velocity command. If a TF lookup fails at the current control tick, the cached command is re-published, preventing the base from stopping during brief connectivity losses.

B. Pre-Planned Body Reconfiguration for Scanning

Once the anticipatory body scanner has localised the patient's anatomy and computed the FAST target set $\mathcal{T}$ in $\mathcal{F}_W$ (Section IV-B), the system must position the Z1 arm to reach each target. Simply keeping Spot at a fixed posture for all targets would force the arm to operate near the edge of its workspace for non-central targets, degrading manipulability and positioning accuracy. TERESA addresses this by pre-planning an optimal body posture for each scanning target before the arm moves.

1) *Workspace sweet spot*: The preferred arm configuration is defined by a *sweet spot* $\mathbf{p}_{\text{sweet}} \in \mathbb{R}^3$ in the arm base frame $\mathcal{F}_0$. This point represents the EE position at which the arm operates with high manipulability, comfortable joint angles, and sufficient reserve to accommodate small patient movements. For the Z1 arm mounted on Spot, $\mathbf{p}_{\text{sweet}}$ is located approximately in front of and slightly above the arm base.

2) *Grid search over body posture*: For each FAST target $\mathbf{p}_i^*$ expressed in the odometry frame $\mathcal{F}_W$, the system searches over a discrete set of feasible body postures. Let $\mathcal{H} = \{h_1, \dots, h_{N_h}\}$ be a set of body heights and $\mathcal{P} = \{\varphi_1, \dots, \varphi_{N_p}\}$ a set of pitch angles. For each candidate

posture (h, φ) , the position of the arm base frame $\mathcal{F}_0$ in $\mathcal{F}_W$ is computed from the known static transform $\mathcal{F}_B \rightarrow \mathcal{F}_0$ and the current body pose. The target position is then transformed to $\mathcal{F}_0$:

$$\mathbf{p}_i^{\mathcal{F}_0}(h, \varphi) = \mathbf{T}_{\mathcal{F}_W}^{\mathcal{F}_0}(h, \varphi) \mathbf{p}_i^*. \quad (13)$$

The score for each posture is the negative Euclidean distance from the transformed target to the sweet spot:

$$s_i(h, \varphi) = -\|\mathbf{p}_i^{\mathcal{F}_0}(h, \varphi) - \mathbf{p}_{\text{sweet}}\|. \quad (14)$$

The optimal posture for target i is $(h_i^*, \varphi_i^*) = \arg \max_{h \in \mathcal{H}, \varphi \in \mathcal{P}} s_i(h, \varphi)$, subject to the posture bounds in (6). The yaw angle θ is never changed during scanning, as it was already optimised during the approach phase.

3) *Inter-target execution*: The body reconfiguration is executed between scanning targets. Before the arm moves to target i , Spot assumes the pre-computed posture (h_i^*, φ_i^*) and is given a settle interval Δt_{settle} to allow the body to stabilise. The target $\mathbf{p}_i^*$ is then re-transformed to $\mathcal{F}_0$ using the live TF tree, reflecting the actual posture achieved. The Z1 FSM receives a “body ready” signal and proceeds with the arm trajectory toward the transformed target. After the arm completes the scan at target i , the cycle repeats for target $i+1$. At the conclusion of the scan, Spot returns to a nominal posture.

Because the entire posture optimisation is computed offline—a single grid search over $\mathcal{H} \times \mathcal{P}$ for each of the N targets, performed immediately after the target set $\mathcal{T}$ is available—it imposes no real-time computational burden and requires no online optimisation during arm motion.

C. Cartesian Impedance Control for Compliant Scanning

The Unitree Z1 arm does not provide native impedance control: its firmware supports only joint position tracking via the Joint Trajectory Controller (JTC) and raw torque commands. To perform ultrasound scanning, however, the probe must maintain gentle and continuous contact with the patient’s skin, adapting to surface curvature and small patient movements. TERESA implements a custom Cartesian impedance controller that runs directly on the Z1’s torque interface.

1) *Cartesian position control*: Let $\mathbf{p}_{EE}(t) \in \mathbb{R}^3$ be the current EE position and $\mathbf{p}_d(t) \in \mathbb{R}^3$ the desired position, updated at 500 Hz from the RealSense surface estimate. The translational position error is:

$$\mathbf{e}_p(t) = \mathbf{p}_d(t) - \mathbf{p}_{EE}(t). \quad (15)$$

The Cartesian control force is computed as:

$$\mathbf{F}_p = \mathbf{K}_p^{\text{eff}} \mathbf{e}_p - \mathbf{K}_d^{\text{eff}} \dot{\mathbf{p}}_{EE} + \mathbf{K}_i \int \mathbf{e}_p dt, \quad (16)$$

where $\dot{\mathbf{p}}_{EE} = \mathbf{J}_{\text{arm}} \dot{\mathbf{q}}$ is the EE translational velocity, and $\mathbf{K}_p^{\text{eff}}, \mathbf{K}_d^{\text{eff}} \in \mathbb{R}^{3 \times 3}$ are effective diagonal gain matrices that adapt to the current arm extension:

$$\mathbf{K}_p^{\text{eff}} = (1 - 0.65\rho) \mathbf{K}_p, \quad \mathbf{K}_d^{\text{eff}} = (1 + \rho) \mathbf{K}_d, \quad (17)$$

where $\rho = \text{clip}(\|\mathbf{p}_{EE}^{xy}\|/r_{\text{max}}, 0, 1) \in [0, 1]$ is the normalised planar reach, and $r_{\text{max}} = 0.6\text{ m}$ is the arm’s nominal planar reach. This adaptive scaling reduces stiffness and

increases damping as the arm extends, improving stability near the workspace boundary. The nominal gains are $\mathbf{K}_p = \text{diag}(250, 250, 300)$, $\mathbf{K}_d = \text{diag}(15, 15, 15)$ in SI units, and $\mathbf{K}_i = \text{diag}(10, 10, 10)$.

2) *Surface-referenced desired position*: When the RealSense surface estimate is available, the desired position is computed relative to the detected patient torso surface. Let $\mathbf{p}_s \in \mathbb{R}^3$ be the surface centroid and $\mathbf{n}_s \in \mathbb{S}^2$ the outward surface normal estimated by the RealSense pipeline. The desired EE position is:

$$\mathbf{p}_d = \mathbf{p}_s + \delta_n \mathbf{n}_s, \quad (18)$$

where $\delta_n = -5\text{ mm}$ is a small offset along the inward normal that ensures light probe contact without excessive force.

3) *Joint torque computation*: The Cartesian force $\mathbf{F}_p$ is projected into joint torques via the arm Jacobian transpose, and combined with joint-space wrist control and dynamic compensation:

$$\boldsymbol{\tau} = \mathbf{J}_{\text{arm}}^\top \tilde{\mathbf{F}}_p + \boldsymbol{\tau}_{\text{wrist}} + \boldsymbol{\tau}_{\text{comp}}, \quad (19)$$

where $\tilde{\mathbf{F}}_p = [\mathbf{F}_p^\top, \mathbf{0}_3^\top]^\top \in \mathbb{R}^6$ pads the translational force with zero moments.

The wrist torque $\boldsymbol{\tau}_{\text{wrist}} \in \mathbb{R}^6$ applies joint-space Proportional-Derivative (PD) control to joints q_4, q_5, q_6 to maintain the probe orientation perpendicular to the skin surface:

$$\tau_{\text{wrist},i} = K_{p,w} (q_{d,i} - q_i) - K_{d,w} \dot{q}_i, \quad i = 4, 5, 6, \quad (20)$$

with $K_{p,w} = 40\text{ N m rad}^{-1}$ and $K_{d,w} = 4\text{ N m s rad}^{-1}$.

The compensation torque $\boldsymbol{\tau}_{\text{comp}} \in \mathbb{R}^6$ comprises gravity and Coriolis terms computed via the Recursive Newton-Euler Algorithm (RNEA) implemented in Pinocchio [42]:

$$\boldsymbol{\tau}_{\text{comp}} = \boldsymbol{\tau}_{\text{gravity}}(\mathbf{q}) + \boldsymbol{\tau}_{\text{Coriolis}}(\mathbf{q}, \dot{\mathbf{q}}). \quad (21)$$

The total torque is saturated to the Z1 hardware limit $\tau_{\text{max}} = 70\text{ N m}$ before being sent to the joint controllers:

$$\boldsymbol{\tau}_{\text{cmd}} = \text{clip}(\boldsymbol{\tau}, -\tau_{\text{max}} \mathbf{1}, \tau_{\text{max}} \mathbf{1}). \quad (22)$$

4) *Controller switching*: The Z1 alternates between two Robot Operating System (ROS) 2 controllers: the JTC for position-controlled approach and homing motions, and the torque controller for impedance-based scanning. A `safe_controller_switch` node exposes two trigger services—`/safe_switch/to_torque` and `/safe_switch/to_jtc`—that the FSM calls before entering and leaving the scanning phase. The JTC is the safe default; the system always returns to it on error. This custom impedance implementation is essential to TERESA’s scanning capability, as the Z1 arm’s native control stack provides no compliant contact mode.

VI. SYSTEM ARCHITECTURE

The TERESA system is organised into four layers—hardware, perception, mission management, and control—communicating over ROS 2 with DDS. Computation is distributed across three units: the Spot onboard computer SpotCore (base control and odometry), a NVIDIA Jetson

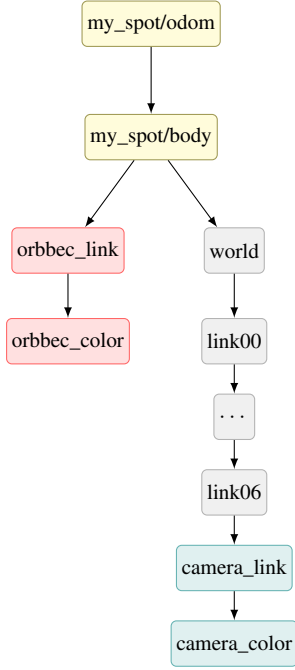


Fig. 1. TF frame tree. $\text{my_spot}/\text{odom} \rightarrow \text{my_spot}/\text{body}$ branches into the Orbbec chain ($\text{orbbeclink} \rightarrow \text{orbbeccolor_optical_frame}$, shown as orbbeccolor) and the Z1 arm chain ($\text{world} \rightarrow \text{link00} \rightarrow \dots \rightarrow \text{link06} \rightarrow \text{cameralink} \rightarrow \text{cameracolor_optical_frame}$, shown as cameracolor).

(perception and arm control), and an external PC (mission coordination and WBC). All components described in this section are available as open-source software.

A. Hardware Platform

The **Boston Dynamics Spot** quadruped provides mobile navigation on unstructured terrain. It is commanded via its onboard ROS 2 driver (`spot_ros2`) through velocity commands $[v_x, \omega_z]$ on the `/cmd_vel` topic and body posture commands on the `/body_pose` topic. The **Unitree Z1** is a six-DOF serial manipulator with a payload of 2 kg, mounted rigidly on Spot’s back. It is controlled at the torque level through a JTC interface running on the Jetson. The **ultrasound probe** is attached to the Z1 flange ($\mathcal{F}_{EE}$).

Two depth cameras provide complementary coverage. An **Orbbec Femto Bolt RGBD** camera is mounted on Spot’s body and provides a wide field of view for patient detection at distances up to 3 m. An **Intel RealSense D435 RGBD** camera is mounted at the Z1 EE and provides close-range depth and colour for surface estimation and torso tracking during scanning.

B. Frame Tree

The TF tree defines the spatial relationships between all system components. Fig. 1 illustrates the frame hierarchy:

The key properties of this tree are:

- $\text{my_spot}/\text{odom}$ is world-fixed, published by SpotCore odometry.
- $\text{my_spot}/\text{body}$ moves with the base (height, pitch, and yaw are set via `/body_pose`).

- $\text{body} \rightarrow \text{orbbeclink}$ and $\text{body} \rightarrow \text{world}$ are static transforms representing the calibrated Orbbec and Z1 mount positions.
- $\text{world} \rightarrow \text{link00}$ is a fixed joint in the Z1 URDF, published by `robot_state_publisher`. The Z1 arm chain $\text{link00} \rightarrow \dots \rightarrow \text{link06}$ follows.
- $\text{link06} \rightarrow \text{cameralink} \rightarrow \text{cameracolor_optical_frame}$ defines the RealSense pose on the EE.

Because the $\text{body} \rightarrow \text{world}$ transform is static, the entire Z1 kinematic tree moves rigidly with Spot, and any change in body posture automatically propagates to the arm base frame $\mathcal{F}_0$.

C. Perception Pipeline

1) *Wide-field patient detection (Orbbec)*: During the SEARCHING and approach phases, YOLO11-pose [43] runs on the Orbbec RGBD stream and detects the patient’s skeleton. A posture classifier labels the patient’s pose (standing, sitting, lying) and outputs a confidence score $c \in [0, 1]$. A laying-human detector computes the approach point as the torso centroid of the four Common Objects in Context (COCO) keypoints (shoulders and hips), back-projected to 3D via the calibrated depth map. It also publishes the torso centroid position—the `body_center`—as a separate `PoseStamped` message, used by the WBC look-at controller to orient the EE camera toward the patient’s body during the pre-approach phase. The approach point is expressed in the camera optical frame; the TF tree transforms it to $\mathcal{F}_W$ for target locking (Section IV-A).

2) *Close-range torso tracking (RealSense)*: Once the mission transitions to the approach sequence, the EE-mounted RealSense camera becomes active. A YOLO-based torso tracker with a six-state Kalman Filter (KF) (position and velocity) publishes a filtered torso estimate $\hat{\mathbf{p}}(t)$ and a tracker state. Three tracking states are relevant during the approach: ESTIMATING (the Kalman filter is converging on at least three torso keypoints with sufficient confidence), LOCKED (the estimate has stabilised with low spatial variance), and GUIDING (a lenient mode used during active search where any partial skeleton detection may trigger the semi-locking phase). A surface estimation node fits a plane to the depth point cloud in a region of interest around the detected torso, publishing the surface centroid $\mathbf{p}_s$ and outward normal $\mathbf{n}_s$ as a `PoseStamped` message.

3) *Sensor handoff strategy*: The transition from Orbbec to RealSense is not a hard switch but a confidence-driven handoff. During PRE_APPROACH the coordinator requires several consecutive readings from the RealSense tracker in either the ESTIMATING or LOCKED state before advancing to the approach phase. This ensures that the close-range sensor has reliably acquired the patient before the Orbbec loses the target at short range. If the RealSense fails to confirm within a timeout, the system still proceeds, falling back to the odometry target acquired during the locking phase.

TABLE I

COMPONENT ACTIVATION BY MISSION PHASE. ● = ACTIVE, ○ = IDLE, — = NOT RELEVANT.

Phase	Spot	Arm	Orbbec	RealSense	Body Scanner
WAITING_TF	○	○	○	○	—
IDLE	○	○	○	○	—
SEARCHING	●	●	●	●	—
SEMI_LOCKING	●	●	●	●	—
LOCKING	○	●	●	○	—
PRE_APPROACH	●	●	○	●	○
APPROACHING	●	●	○	●	●
SCANNING	●	●	—	●	●
WS_EXTENSION	●	●	—	●	—

D. Mission Finite State Machine

The mission is orchestrated by a nine-state FSM implemented in the `wbc_coordinator` node. Fig. 2 shows the state transitions; Table I summarises which components are active in each state.

- **Waiting TF.** The system waits for the `tf_monitor` node to confirm that all TF chains and hardware topics are available. This gate prevents premature activation when SpotCore DDS discovery is still in progress.
- **Idle.** All TF ready. The system waits for a start command from the keyboard operator.
- **Searching.** Spot sweeps through a sequence of body yaw orientations in a coarse search, while the arm executes an `ACTIVE_SEARCH` mode with wide Cartesian poses. Two detection mechanisms run in parallel: Orbbec direct detection of a lying patient and RealSense guidance mode for partial body detection (Section IV-A).
- **Semi-locking.** The RealSense has detected a potential body but Orbbec confirmation is pending. Spot rotates toward the detected torso direction while the arm maintains a WBC look-at orientation. The Orbbec receives a clean observation window to confirm the detection. Upon confirmation, the system transitions to locking; on timeout, it resumes searching from the current orientation.
- **Locking.** The Orbbec has confirmed a lying patient. The arm returns to an elevated home pose, while the coordinator collects a sequence of approach point samples. The transition to pre-approach is gated by both the IK trajectory completion and the collection of the required number of samples.
- **Pre-approach.** Spot straightens to an upright posture. The WBC arm look-at controller orients the RealSense toward the patient's torso centroid (`body_center`, computed from the Orbbec skeleton during locking). The coordinator waits for consecutive `ESTIMATING` or `LOCKED` tracker readings before authorising the approach. A global timeout ensures the mission progresses even if the tracker cannot achieve full lock.
- **Approaching.** The Spot navigator drives toward the approach target while the adaptive anticipatory body scanner (Section IV-B) executes the Cartesian scan grid concurrently with base motion. A global timeout guards against unreachable targets, aborting to idle if Spot fails to reach the handoff distance.

TABLE II

WBC AND MISSION ORCHESTRATION NODES.

Node	Role
<code>wbc_coordinator</code>	Nine-state mission FSM, QualityMonitor, body pose grid search optimisation, dual-sensor lock management.
<code>wbc_qp_controller</code>	Arm WBC look-at via damped pseudo-inverse, adaptive Cartesian scan grid generation, keypoint confidence fusion, FAST target publishing.
<code>wbc_spot_navigator</code>	Simplified approach navigator: rotate → drive → stop toward the odometry goal.
<code>laying_human_detector</code>	Orbbec skeleton-based approach point and torso centroid (<code>body_center</code>) computation for navigation and WBC look-at targeting.
<code>tf_monitor</code>	Continuous monitoring of eight TF chains and three hardware topics; publishes <code>/wbc/tf_ready</code> .
<code>ik_goal_mux</code>	Priority multiplexer: WBC arm goals override Z1 FSM goals when WBC is enabled.

TABLE III

Z1 ARM CONTROL AND SCANNING NODES.

Node	Role
<code>z1_fsm</code>	Arm mission FSM: HOMING → WAITING → CHECKING_WORKSPACE → APPROACHING → FAST cycle.
<code>z1_ik_to_jtc</code>	Pinocchio damped pseudo-inverse IK with smoothstep quintic trajectory generation → JTC action.
<code>safe_controller_switch</code>	Service-based switching between JTC (position) and torque controller (impedance).
<code>impedance_controller_realsense</code>	Cartesian impedance at 500 Hz, adaptive gains, RNEA gravity and Coriolis compensation.
<code>z1_yolo_torso_tracker</code>	RealSense YOLO-based torso tracker with Kalman filter, keypoint confidence reporting, and multi-state tracking (<code>GUIDING</code> , <code>ESTIMATING</code> , <code>LOCKED</code>).
<code>z1_scan_manager</code>	Computation of five FAST target poses from fused torso keypoints.

- **Scanning.** Handoff distance reached. Spot assumes a lowered body posture. The body scanner publishes the FAST target set. The coordinator executes the pre-planned body reconfiguration between targets (Section V), while the Z1 FSM drives the arm through each scanning target.
- **WS ext.** If a target is reported unreachable, Spot executes a micro-step to bring the target back into the workspace. Once confirmed, the FSM returns to scanning.

At any time, a stop command from the keyboard immediately transitions the FSM to idle, bringing the base to a halt and returning the arm to a safe configuration.

E. WBC and Mission Components

The WBC layer comprises the nodes launched by `wbc.launch.py`, listed in Table II.

F. Z1 Control Components

The Z1 arm control stack, listed in Table III, is independent of the WBC layer and can operate in standalone mode for fixed-base scanning.

G. Z1 FSM Integration with WBC Coordinator

The Z1 FSM and the WBC coordinator communicate through a small set of ROS topics to synchronise their state machines.

In standalone mode (no WBC), the Z1 FSM executes its own body scanning phase after homing. When integrated with TERESA, the FSM enters `WAITING` after homing and listens for two conditions: `/wbc/state = 'SCANNING'` (indicating that the approach is complete) and `/z1/fast_ready = True` (indicating that the anticipatory body scanner has

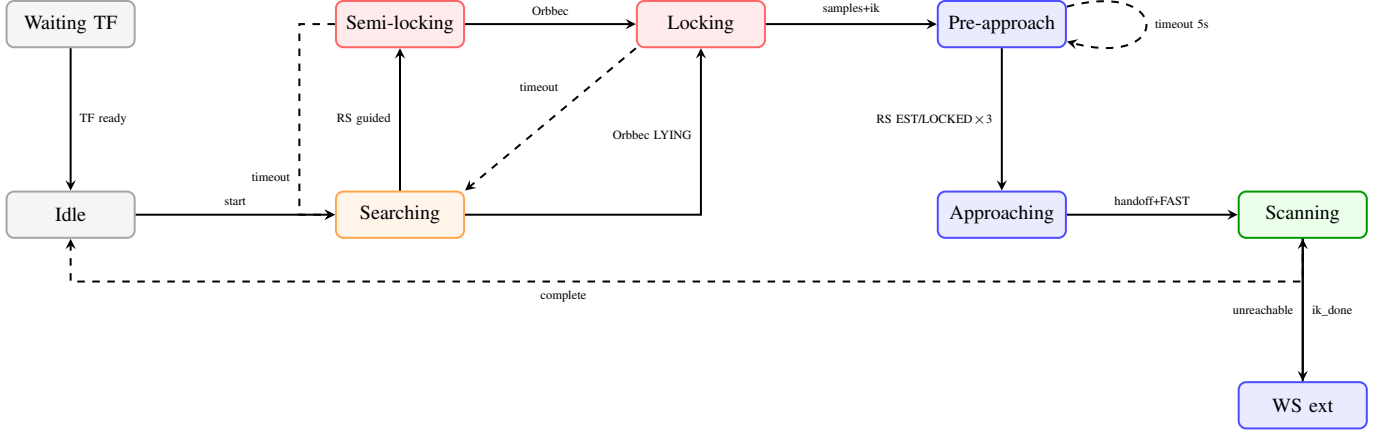


Fig. 2. Mission FSM. Transitions are triggered by perception confidence (dashed) or geometric conditions (solid). WS_EXT = workspace extension.

computed the FAST target set). If both are received, the FSM skips its internal BODY_SCANNING phase entirely, avoiding redundant scanning.

During the FAST cycle, the WBC coordinator optimises Spot’s body posture for each target and broadcasts `/wbc/body_ready`. The Z1 FSM waits for this signal before executing the approach to the next target. After each target is reached, the FSM publishes `/z1/next_point_idx` to request the next body posture optimisation. This synchronisation loop continues for all N targets.

H. Operational Flow

The full mission is launched from five terminal sessions:

- 1) `teresa_core.launch.py` — Spot driver, Orbbec driver, RealSense driver, Z1 bringup, static TF transforms, `tf_monitor`.
- 2) `teresa_perception.launch.py` — Orbbec YOLO perception nodes and RealSense YOLO torso tracker.
- 3) `z1_control.launch.py` — Z1 arm control stack (IK, FSM, controller switching, impedance).
- 4) `wbc.launch.py` — WBC coordinator, QP controller, spot navigator, IK goal multiplexer.
- 5) `wbc_keyboard_node` — Operator keyboard interface (start, stop, emergency return).

Upon launch, the `tf_monitor` verifies all TF chains and hardware topics. When ready, the keyboard operator presses the start key, and the mission proceeds through the FSM states automatically until all scanning targets are completed or an emergency stop is triggered.

VII. EXPERIMENTAL SETUP

[TODO: Write experimental setup]

VIII. RESULTS AND DISCUSSION

[TODO: Write results and discussion]

IX. CONCLUSION

[TODO: Write conclusion]

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